

Compression and Segmentation for Energy-Constrained Wireless Multimedia Sensor Nodes: A Comparative Study and Energy-Aware Prototypes

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Abstract. Wireless Multimedia Sensor Networks (WMSNs) must capture and transmit image data under tight energy, memory, and bandwidth budgets. In such platforms, radio transmission dominates energy consumption, which makes compression and lightweight segmentation central design decisions rather than secondary optimizations. This paper makes two contributions. First, it compares five compression algorithms (Huffman, LZW, JPEG 2000 ROI, Slepian-Wolf, and Wyner-Ziv) and six segmentation algorithms (Otsu, adaptive thresholding, K-means, region growing, split-and-merge, and MobileNet-Seg) on 871 plant-leaf images using image quality, compression efficiency, execution time, CPU load, and estimated energy. Wyner-Ziv provides the best compression result, reaching a $12\times$ compression ratio with a 0.023s encoding time, while Otsu delivers the best segmentation trade-off, with a global score of 95.66/100 at 0.0016s and a $3.5\times$ higher IoU than MobileNet-Seg. Second, we introduce two node-oriented pipelines: WZ-OSEG, which combines Wyner-Ziv distributed coding with hybrid Otsu-HSV segmentation, and ADRES, which applies region-aware compression through adaptive energy profiles. For 640×480 RGB images in the prototype setting, these methods keep encoder-side complexity linear and reduce the transmitted data to 5–16 KB range, depending on the selected profile. An energy-gating mechanism suppresses 70% of redundant transmissions with 98% accuracy, extending node lifetime by $3.3\times$. Overall, the recommended pipeline yields up to 90% energy savings compared with continuous JPEG transmission.

Keywords: Wireless multimedia sensor networks · Image compression · Image segmentation · Wyner-Ziv coding · Asymmetric compression · Energy efficiency · Otsu thresholding

1 Introduction

Wireless Multimedia Sensor Networks are increasingly used in precision agriculture, environmental monitoring, industrial inspection, and other applications in which battery-powered nodes must acquire images, process them locally, and transmit only what is worth sending [1]. The challenge is not image acquisition itself, but sustainable communication. In low-power camera nodes, the radio transceiver remains the dominant energy sink [15], so every unnecessary byte sent over the air directly reduces system lifetime.

Most prior work studies compression and segmentation separately. Compression papers often optimize rate-distortion performance without asking whether the encoder can realistically run on a constrained node. Segmentation papers, in turn, may report mask quality but say little about the communication savings that segmentation can unlock when only regions of interest are transmitted. What is still relatively rare is a joint, deployment-oriented evaluation that compares lightweight candidates under one protocol and then turns the strongest ones into operational node-side pipelines.

This paper addresses that gap through two contributions that are deliberately aligned with the needs of embedded WMSNs. First, we provide a comparative evaluation of five compression algorithms and six segmentation algorithms on a common dataset of 871 plant-leaf images. Second, based on the best-performing building blocks, we design two energy-aware prototype pipelines—WZ-OSEG and ADRES—and combine them with an energy-gating mechanism that suppresses redundant transmissions before they reach the radio.

The rest of the paper is organized as follows. Section 2 reviews the most relevant work on compression and segmentation for WMSNs. Section 3 summarises the evaluation protocol and the benchmark results that motivate our design choices. Section 4 presents the proposed pipelines. Section 5 discusses the main findings, deployment implications, and current limitations. Section 6 concludes the paper.

2 Related Work

The literature on image compression for WMSNs broadly follows two directions. The first includes classical lossless or near-lossless methods such as Huffman [2] and LZW coding [3]. These techniques are simple, but they usually provide limited byte reduction on natural images, which restricts their value when radio transmission dominates energy consumption [1]. The second direction includes transform-based codecs such as JPEG [4] and JPEG 2000 [5], which reach much better compression rates, but often move too much complexity to the encoder side for long-lived sensor nodes.

Distributed source coding offers a more attractive compromise for embedded sensing. Slepian-Wolf coding [6] and, in particular, Wyner-Ziv coding [7] shift a larger share of the reconstruction burden to the decoder, which better matches the asymmetry between weak sensor nodes and more capable sinks or gateways.

The applicability of Wyner-Ziv coding to video and correlated image streams was demonstrated by Aaron et al. [8] and Girod et al. [9]. This property makes Wyner-Ziv especially relevant in scenarios where images are temporally correlated, as is often the case in crop monitoring or static environmental observation.

On the segmentation side, lightweight thresholding methods remain highly competitive when the scene structure is relatively regular. Otsu thresholding [10] is still appealing because it is simple, fast, and effective on images with separable foreground and background distributions. Adaptive thresholding [11] is often more robust to illumination changes, but at a somewhat higher computational cost. By contrast, deep models such as MobileNet-Seg [12] can be accurate in well-matched domains, yet they require more compute and usually need domain-specific training before they become convincing candidates for on-node deployment.

Taken together, these observations suggest that the best WMSN pipeline is unlikely to come from a single sophisticated component. Instead, it should emerge from a carefully balanced combination of lightweight segmentation, communication-aware compression, and transmission suppression. This is the perspective adopted in the present work.

3 Comparative Evaluation and Design Rationale

Our evaluation follows a practical question: which combination of compression and segmentation is most credible for an energy-constrained camera node? To answer it, we benchmarked five compression algorithms: Huffman [2], LZW [3], JPEG 2000 ROI [5], Slepian-Wolf [6], and Wyner-Ziv [7], and six segmentation algorithms: Otsu [10], adaptive thresholding [11], K-means, region growing, split-and-merge, and MobileNet-Seg [12].

The comparative study used 871 plant-leaf images for the algorithmic evaluation. The prototype stage then relied on 50 maize-plant images at 640×480 resolution (about 921 KB in raw RGB form), while a dedicated 20-image-pair sequence was used to validate the energy gate. Evaluation metrics covered image quality (SSIM [13] and PSNR), compression efficiency, execution time, CPU usage, segmentation quality (IoU and Dice), and estimated energy. In line with the WMSN setting, the energy model distinguishes processing costs from transmission cost:

$$E_{\text{total}} = E_{\text{CPU}} + E_{\text{RAM}} + E_{\text{TX}} = 0.0054 t_{\text{CPU}} + 0.00001 M_{\text{RAM}} + 0.18 B_{\text{compressed}} \quad [\text{mJ}] \quad (1)$$

where the 0.18 mJ/byte transmission coefficient is derived from IEEE 802.15.4 radio parameters, with the latter acting as the main driver of node lifetime [15].

3.1 Compression Benchmarks

The compression benchmark highlights an important practical lesson: perfect reconstruction alone is not enough to make an algorithm useful on a radio-dominated platform. Huffman [2] and LZW [3] preserve the source well, but

their byte reduction is too small to justify their use as primary image codecs in WMSNs. JPEG 2000 ROI [5] remains attractive when region quality must be strongly protected, but its encoder is heavier than what many embedded nodes can afford for long missions.

Slepian-Wolf coding [6] offers a better compromise when near-lossless behaviour is important. However, the strongest overall result comes from Wyner-Ziv coding [7–9], which achieves the best compression performance in the study, with a $12\times$ compression ratio and the lowest encoding time at 0.023 s. This makes Wyner-Ziv the most suitable compression backbone for a node-oriented design in which encoder simplicity matters as much as raw rate reduction.

Table 1: Compression benchmark summary.

Algorithm	Main observation	WMSN relevance
Huffman [2]	Lossless reconstruction, but limited byte reduction on natural images.	Too weak as a primary image codec when radio cost dominates.
LZW [3]	Simple and reliable, yet still offers insufficient compression for image-heavy sensing.	Useful only in niche cases where strict simplicity matters more than savings.
JPEG 2000 ROI [5]	Protects the region quality well, but the encoder is comparatively heavier.	Relevant when ROI fidelity is critical, and the node budget is less restrictive.
Slepian-Wolf [6]	Near-lossless behaviour with an attractive encoder/decoder asymmetry.	A strong choice when fidelity is prioritized over maximum compression.
Wyner-Ziv [7]	Best compression result in the benchmark ($12\times$) and lowest encoding time (0.023 s).	Best overall compression backbone for energy-limited node-side deployment.

3.2 Segmentation Benchmarks

The segmentation benchmark leads to an equally clear conclusion. Otsu thresholding [10] provides the best overall trade-off on the agricultural dataset, with a global score of 95.66/100 and an execution time of 0.0016 s. In addition to its speed, it outperforms MobileNet-Seg [12] by a factor of 3.5 in IoU (0.537 vs 0.150), which shows that lightweight classical methods still deserve serious attention in constrained deployments.

Adaptive thresholding [11] remains a useful alternative when illumination is less uniform, but it does not improve enough over Otsu to offset its additional cost in the present setting. K-means and split-and-merge can produce acceptable masks, yet their runtimes are harder to justify on a node that must stay energy-frugal. Region growing is unstable on complex leaf scenes, and MobileNet-Seg [12]

underperforms on this dataset, which likely reflects insufficient domain adaptation rather than a fundamental limitation of neural segmentation.

Table 2: Segmentation benchmark summary.

Algorithm	Observed behaviour	Deployment reading
Otsu [10]	Best global score (95.66/100), Best overall trade-off for em-0.0016s runtime, and 3.5× bedded agricultural monitor-higher IoU than MobileNet-ing. Seg.	
Adaptive thresh. [11]	Robust under illumination changes, but slightly more unstable. costly than Otsu.	Good fallback when lighting is
K-means	Reasonable mask quality, but Hard to justify on energy-much slower than thresholding constrained nodes. methods.	
Region growing	Sensitive to clutter and local variability on leaf scenes.	Limited reliability in the tested setting.
Split-and-merge	Competitive masks, but a higher runtime reduces its practical appeal.	Possible off-node or gateway-side option.
MobileNet-Seg [12]	Underperformed on the current dataset, likely due to insufficient domain adaptation.	Not yet persuasive for on-node training.

4 Proposed Energy-Aware Pipelines

The benchmark results suggest a straightforward design principle: keep node-side processing linear, preserve region information when it matters, and avoid sending full images unless the scene has genuinely changed. Based on this principle, we designed two prototype pipelines.

4.1 WZ-OSEG

WZ-OSEG combines three ideas in one node-oriented pipeline. First, it performs a lightweight change analysis using global visual descriptors—luminance mean, luminance variance, histogram entropy, and green-pixel fraction—so that nearly identical frames do not trigger unnecessary transmissions. The change decision uses the normalized ℓ_2 distance between successive descriptor vectors [1]:

$$d(\mathbf{ind}_t, \mathbf{ind}_{t-1}) = \sqrt{\sum_{i=1}^d w_i \left(\frac{ind_{t,i} - ind_{t-1,i}}{\sigma_i} \right)^2} \quad (2)$$

If $d < \tau_{\text{chg}}$, only a metadata packet (< 100 bytes) is transmitted; otherwise, the full pipeline is activated.

Second, it extracts a region of interest through a hybrid Otsu-HSV segmentation stage, which well-suited to vegetation images where luminance and hue both carry discriminative information. The greyscale conversion follows ITU-R BT.601:

$$L = 0.299 R + 0.587 G + 0.114 B \quad (3)$$

The Otsu threshold [10] maximises inter-class variance:

$$\sigma_B^2(t) = \omega_0(t) \cdot \omega_1(t) \cdot [\mu_0(t) - \mu_1(t)]^2, \quad t^* = \arg \max_{t \in [0, L-1]} \sigma_B^2(t) \quad (4)$$

A hue mask M_{hue} selects pixels with $H \in [30^\circ, 90^\circ]$ (vegetation hue range in HSV space). The final ROI mask is the logical conjunction:

$$M_{\text{ROI}}(x, y) = M_{\text{Otsu}}(x, y) \wedge M_{\text{hue}}(x, y) \quad (5)$$

Third, it applies Wyner-Ziv distributed coding [7] so that most of the reconstruction complexity is transferred to the sink rather than being paid on the sensor node. The luminance channel is block-DCT transformed (8×8 blocks), 21 low-frequency coefficients are retained ($u + v < 6$), and uniform scalar quantisation with step Δ produces indices $q_i = \lfloor x_i / \Delta \rfloor$. LDPC syndrome generation then forms the transmitted bitstream:

$$\mathbf{s} = H \cdot \mathbf{q} \bmod 2 \quad (6)$$

where H is the parity-check matrix of the chosen LDPC code. For a 640×480 image at coding rate $R = 0.6$, the resulting communication volume is:

$$V_{\text{comm}} = \frac{(1 - R) \cdot n}{8} + 170 \text{ bytes} \approx 15.2 \text{ KB} \quad (\approx 60 \times \text{reduction vs. raw RGB}) \quad (7)$$

The resulting encoder remains linear in the number of pixels and is therefore compatible with constrained hardware. Its main strength is that it uses temporal correlation without overloading the node, which makes it especially relevant for periodic monitoring scenarios.

4.2 ADRES

ADRES is designed for situations where frame independence and deterministic latency matters more than temporal coding gains. Instead of relying on inter-frame correlation, it applies spatially differentiated compression: regions of interest are preserved with a higher-quality profile, while background areas are encoded more aggressively. This asymmetry is controlled through adaptive energy profiles, allowing the system to move between two operating modes:

- **Quality (Q):** ROI sub-sampled $\times 2$, quantised with step 1.5; background sub-sampled $\times 4$, step 2. $V_{\text{comm}} \approx 16 \text{ KB}$.

- **Economy (E):** ROI sub-sampled $\times 4$, step 2.0; background sub-sampled $\times 8$, step 3.0. $V_{\text{comm}} \approx 5 \text{ KB}$ ($\approx 180\times$ reduction vs. raw RGB).

Node-side processing in ADRES follows a hue-based segmentation stage (dual-interval HSV mask with morphological refinement: erosion 3×3 , dilation 5×5), a bicubic sub-sampling step applied independently to ROI and background, and a PNG lossless compression of the quantized data. An RLE-compressed binary mask is transmitted as compact metadata. Sink-side decoding applies bicubic or SRCNN super-resolution for ROI up-sampling, bilateral filtering on the ROI [16], Gaussian smoothing on the background, alpha-blending fusion at boundaries (feathering radius 5 px), and guided filtering for global coherence [17].

Like WZ-OSEG, ADRES keeps node-side complexity linear. Its main advantage is operational predictability: each image is handled independently, which simplifies random access and makes the pipeline easier to deploy in scenes with faster changes or less stable temporal structure.

4.3 Energy-Gating Mechanism

The energy gate sits before compression and acts as the first line of defence against wasted radio usage. It evaluates scene change through three lightweight criteria: (1) histogram ℓ_1 distance greater than 0.38, (2) mean RGB channel difference greater than 6.0, and (3) fraction of changed 32×32 blocks greater than 18%. A transmission is triggered only when at least two of these three criteria indicate meaningful novelty.

This mechanism suppresses 70% of redundant transmissions with 98% accuracy—validated on 190 image pairs—and extends node lifetime by a factor of 3.3. More importantly, it changes the design logic of the whole system: the best energy savings do not come only from better coding, but from combining coding with selective transmission. In the recommended deployment pipeline, this combination yields up to 90% energy savings relative to continuous JPEG [4] transmission (753 mJ/day without gate $\rightarrow$ 226 mJ/day with gate).

Table 3: Deployment-oriented view of the proposed pipeline.

Component	Key role	Main outcome
WZ-OSEG	Hybrid Otsu-HSV ROI extraction followed by Wyner-Ziv [7] coding.	Linear encoder-side complexity and compact transmission for temporally correlated scenes.
ADRES	Independent-frame, region-aware compression with adaptive energy profiles.	Deterministic latency and flexible quality/energy trade-off.
Energy gate	Suppresses transmissions when scene variation is not avoided with significant.	70% redundant transmissions avoided with 98% accuracy; 3.3× lifetime extension.
Recommended pipeline	Energy gate combined with the most suitable codec for the continuous scene.	Up to 90% energy savings over the continuous JPEG [4] transmission.

5 Results, Implications, and Limitations

5.1 Key Findings

The first major outcome of this study is methodological. Compression and segmentation should not be optimized in isolation in WMSNs. A fast segmentation method with no impact on the transmitted volume is of limited practical value, just as a high-fidelity codec is of limited use if its encoder is too heavy for an embedded node. The most convincing designs are those that jointly reduce transmitted bytes, preserve useful visual structure, and keep node-side computation predictable.

The second major outcome is architectural. Wyner-Ziv [7] and Otsu [10] emerge as the most credible low-complexity building blocks in the present agricultural setting. WZ-OSEG is therefore the better choice for continuous or quasi-static monitoring, where temporal correlation can be exploited. ADRES is the more robust option when independent-frame processing, deterministic latency, or easier random access is required. The two methods are not competing so much as covering different deployment regimes.

The prototype results also reinforce a central lesson about energy in camera sensor nodes [15]: the radio dominates. This is why the most meaningful gains come from a combination of strong compression and avoided transmissions. In practical terms, reducing a 640×480 RGB image to the 5–16 KB range and suppressing redundant frames is far more influential for node lifetime than shaving a few milliseconds from local processing alone. The energy model (Eq. 1) confirms this quantitatively: E_{TX} accounts for 99.7% of total node energy at the prototype scale.

5.2 Prototype Results Summary

Table 4 reports mean results over 50 maize-plant images.

Table 4: Mean prototype results over 50 maize-plant images ($\uparrow$ higher is better, $\downarrow$ lower is better).

Algorithm	SSIM $\uparrow$	PSNR $\uparrow$	Ratio $\uparrow$	CPU (ms) $\downarrow$	RAM (MB) $\downarrow$	Energy (mJ) $\downarrow$
JPEG [4] (ref.)	0.970	33.97	11.3 $\times$	60.8	5.4	15.7
JPEG 2000 ROI [5] (ref.)	0.983	36.48	10.0 $\times$	107.2	17.0	17.4
WZ-OSEG (Proto 1)	0.866	19.53	2.6 $\times$	40.5	4.1	63.9
ADRES (Proto 2)	0.882	24.79	0.5 $\times$	18.6	4.2	330.7

The compression gap between the prototypes and the industrial references reflects two known design flaws. For WZ-OSEG, the fixed-size DCT serialization transmits all 21 retained coefficients per block regardless of their magnitude, without any entropy coding; adding RLE + Huffman over the coefficient stream is expected to reduce size by 3–4 $\times$ and bring the compression ratio to 8–10 $\times$. For ADRES, the sparse positional format (7 bytes per pixel) generates more overhead than the sub-sampling saves, replacing it with dense sequential ROI storage followed by a JPEG-compressed background is expected to yield ≈ 70 KB per image (ratio $> 10\times$).

5.3 Limitations

The present study has clear limits. The evaluation is centred on agricultural imagery and software prototypes rather than on a full hardware-in-the-loop deployment. The behaviour of MobileNet-Seg [12] should also not be overgeneralized, because lightweight neural models usually need better domain-specific training data to perform well. Future work will therefore focus on hardware validation on embedded platforms such as STM32H7 or ESP32-CAM, improved entropy coding for WZ-OSEG, more compact serialization for ADRES, and extension to video sequences exploiting H.264-style inter-frame prediction.

6 Conclusion

This paper presented a deployment-oriented study of compression and segmentation for WMSNs. The comparative evaluation shows that Wyner-Ziv [7] offers the strongest compression-oriented compromise for constrained nodes, while Otsu [10] remains the most convincing segmentation method on the tested agricultural dataset. Building on these findings, the proposed WZ-OSEG and ADRES pipelines keep encoder-side complexity linear and reduce transmitted

image volume to the 5–16 KB range for 640×480 RGB inputs in the prototype setting.

When combined with the proposed energy gate, the overall pipeline suppresses redundant transmissions with 98% accuracy, extends node lifetime by $3.3\times$, and achieves up to 90% energy savings compared with continuous JPEG [4] transmission. These results suggest that practical progress in WMSNs will come less from isolated algorithmic sophistication than from carefully engineered combinations of lightweight vision, asymmetric compression, and communication-aware decision making.

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